K-Means

Objetivo:
$\min_{\{r_{nk}\},\{\boldsymbol{\mu}_k\}} J = \sum_n \sum_k r_{nk}\|\mathbf{x}_n - \boldsymbol{\mu}_k\|^2$
$r_{nk}\in\{0,1\}$, $\sum_k r_{nk}=1$ (asignación *hard*)

Inicialización $\boldsymbol{\mu}_k$ (random o K-Means++)
≈**E-Step**: $z_n^* \leftarrow \arg\min_k \|\mathbf{x}_n - \boldsymbol{\mu}_k\|^2$
≈**M-Step**: $\boldsymbol{\mu}_k \leftarrow \frac{1}{N_k}\sum_{n:\, z_n^*=k} \mathbf{x}_n$
K-Means++: $p(\boldsymbol{\mu}_{t+1} = \mathbf{x}_n) = \frac{d_t(\mathbf{x}_n)}{\sum_{n'} d_t(\mathbf{x}_{n'})}$
$d_t(\mathbf{x}) = \min_{k=1}^t \|\mathbf{x}-\boldsymbol{\mu}_k\|_2^2$
Límite de GMM: $\boldsymbol{\Sigma}_k = \varepsilon\mathbf{I}$, $\varepsilon\to 0 \Rightarrow$ asignación hard (gaussianas infinitamente concentradas).

Gaussian Mixture Model (GMM)

$p(\mathbf{x}|\mathbf{w}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)$; $\sum_k \pi_k = 1$
Latente: $z\sim \mathrm{Cat}(\boldsymbol{\pi})$; $\mathbf{x}|z=k \sim \mathcal{N}(\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)$
$\ln p(\mathbf{X}|\mathbf{w}) = \sum_n \ln\sum_k \pi_k\mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)$

E-step (responsabilidades, soft clust.):
$$r_{nk} = p(\mathbf{z}=k\mid x_n,\boldsymbol{\theta}) = \frac{\pi_k\mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)}{\sum_{k'}\pi_{k'}\mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_{k'},\boldsymbol{\Sigma}_{k'})}$$
M-step:
$N_k = \sum_n r_{nk}$; $\hat\pi_k = \frac{N_k}{N}$; $\hat{\boldsymbol{\mu}}_k = \frac{1}{N_k}\sum_n r_{nk}\mathbf{x}_n$
$\hat{\boldsymbol{\Sigma}}_k = \frac{1}{N_k}\sum_n r_{nk}(\mathbf{x}_n-\hat{\boldsymbol{\mu}}_k)(\mathbf{x}_n-\hat{\boldsymbol{\mu}}_k)^\top$
Asignación (hard clust.): $\mathbf{z}_n = \arg\max_k r_{nk}$
Frontera GMM: $r_{nk} = r_{nk'}$, lineal si $\boldsymbol{\Sigma}_k = \boldsymbol{\Sigma}\,\forall k$.

EM & ELBO

Goal: $\max_{\mathbf{w}} \ln p(\mathbf{x}|\mathbf{w}) = \max_{\mathbf{w}}\ln\sum_{\mathbf{z}} p(\mathbf{x},\mathbf{z}|\mathbf{w})$
Asumiendo $q(\mathbf{z})$:
$$\ln p(\mathbf{x}|\mathbf{w}) = \underbrace{\sum_{\mathbf{z}} q(\mathbf{z})\ln\frac{p(\mathbf{x},\mathbf{z}|\mathbf{w})}{q(\mathbf{z})}}_{\text{Ł}(\mathbf{w},q)} + \underbrace{\mathrm{KL}(q\|p(\mathbf{z}|\mathbf{x},\mathbf{w}))}_{\ge 0}$$
$\therefore\ \ln p(\mathbf{x}|\mathbf{w}) \ge \text{Ł}(\mathbf{w},q)$ (**ELBO lower bound**)
E-step: Maximizar Ł respecto q
$q^* = p(\mathbf{z}|\mathbf{x},\mathbf{w}^{t-1}) \Rightarrow$ KL= 0, Ł $=\ln p$
M-step: Maximizar Ł respecto $\mathbf{w}$
$\mathbf{w}^t = \arg\max_{\mathbf{w}} \mathbb{E}_{q(\mathbf{z})}[\ln p(\mathbf{x},\mathbf{z}|\mathbf{w})]$
Convergencia: $\ln p(\mathbf{x}|\mathbf{w}^t)\ge \ln p(\mathbf{x}|\mathbf{w}^{t-1})$
(no-decreciente; converge al menos a máx. local)
ELBO expandido: Ł $= \mathbb{E}_q[\ln p(\mathbf{x},\mathbf{z}|\mathbf{w})] + \mathbb{H}(q)$

Demo: $\mathcal{L}$ VAE

Para cualquier $q(\mathbf{z})$, multiplicar y dividir por $q(\mathbf{z})$:
$$\ln p(\mathbf{x}|\mathbf{w}) = \ln\sum_{\mathbf{z}} p(\mathbf{x},\mathbf{z}|\mathbf{w})$$
$$= \ln\sum_{\mathbf{z}} q(\mathbf{z})\frac{p(\mathbf{x},\mathbf{z}|\mathbf{w})}{q(\mathbf{z})}$$
Separar en dos términos usando $\ln\frac{p(\mathbf{x},\mathbf{z})}{q} = \ln\frac{p(\mathbf{x}|\mathbf{w})p(\mathbf{z}|\mathbf{x},\mathbf{w})}{q}$:
$$= \underbrace{\sum_{\mathbf{z}} q(\mathbf{z})\ln\frac{p(\mathbf{x},\mathbf{z}|\mathbf{w})}{q(\mathbf{z})}}_{\mathcal{L}(q,\mathbf{w})} + \underbrace{\sum_{\mathbf{z}} q(\mathbf{z})\ln\frac{q(\mathbf{z})}{p(\mathbf{z}|\mathbf{x},\mathbf{w})}}_{\mathrm{KL}(q\|p(\mathbf{z}|\mathbf{x},\mathbf{w}))}$$
Como KL ≥ 0: $\ln p(\mathbf{x}|\mathbf{w}) \ge \mathcal{L}(q,\mathbf{w})$ ($\mathcal{L}$ es cota inf.)

Demo: max varianza equiv min error

$\mathrm{Var}[z] = \frac1N\sum_i z_i^2 = \frac1N\sum_i(\mathbf{x}_i^\top\hat{\mathbf{v}})^2 = \hat{\mathbf{v}}^\top\hat{\boldsymbol{\Sigma}}\hat{\mathbf{v}}$
MSE $= \frac1N\sum_i\|\mathbf{x}_i\|^2 - \hat{\mathbf{v}}^\top\hat{\boldsymbol{\Sigma}}\hat{\mathbf{v}}$
$=$ const $-$ Var$[z]$
$\Rightarrow$ min MSE $\Leftrightarrow$ max Var$[z]$. $\square$

Demo: mas K implica log-lik no decrece

Sea $\mathbf{w}_K^*$ óptimo para K clusters. Construir modelo con $K+1$ clusters tomando $\mathbf{w}_{K+1}$ con $\pi_{K+1}=0$ (degenerar nuevo cluster).
$\Rightarrow p_{K+1}(\mathbf{X}|\mathbf{w}_{K+1}) = p_K(\mathbf{X}|\mathbf{w}_K^*)$
$\Rightarrow \ln p(\mathbf{X}|\mathbf{w}_{K+1}^*) \ge \ln p(\mathbf{X}|\mathbf{w}_{K+1}) = \ln p(\mathbf{X}|\mathbf{w}_K^*)$
(el óptimo sobre $K+1$ no puede ser peor que incluir el caso K). $\square$

PCA

Centrar: $\tilde{\mathbf{X}}$; $\hat{\boldsymbol{\Sigma}} = \frac1N\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}$
Eigendecomposición: $\hat{\boldsymbol{\Sigma}}\mathbf{u}_l = \lambda_l\mathbf{u}_l$;
$\mathbf{U}_L = [\mathbf{u}_1,\dots,\mathbf{u}_L]\in\mathbb{R}^{D\times L}$
Encode: $\mathbf{z}_n = \mathbf{U}_L^\top\tilde{\mathbf{x}}_n$; **Decode**: $\hat{\mathbf{x}}_n = \mathbf{U}_L\mathbf{z}_n + \hat{\boldsymbol{\mu}}$
Error recons.: $\frac1N\|\tilde{\mathbf{X}} - \tilde{\mathbf{X}}\mathbf{U}_L\mathbf{U}_L^\top\|_F^2 = \sum_{l=L+1}^D\lambda_l$
Var. explicada: VE$(L) = (\sum_{l=1}^L\lambda_l)/(\sum_{l=1}^D\lambda_l)$
Vía SVD: $\tilde{\mathbf{X}} = \mathbf{U}_{\mathbf{X}}\mathbf{S}\mathbf{V}^\top$; PC's = cols de $\mathbf{V}$; $\lambda_l = s_l^2/N$
Var de z: Var$[z_l] = \lambda_l$ (coordenadas decorreladas)

Autoencoders

AE determinístico: $\mathbf{z} = f_\phi(\mathbf{x})$, $\hat{\mathbf{x}} = g_\theta(\mathbf{z})$, $L\ll D$
$\mathcal{L}_{\text{AE}} = \frac1N\sum_n\|\mathbf{x}_n - g_\theta(f_\phi(\mathbf{x}_n))\|^2$
AE lineal ≡ PCA (mismo subespacio); con no-linealidades → reducción no-lineal.

VAE: prior $p(\mathbf{z}) = \mathcal{N}(\mathbf{0},\mathbf{I})$
Encoder: $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z}|\boldsymbol{\mu}_\phi(\mathbf{x}),\mathrm{diag}(\boldsymbol{\sigma}_\phi^2(\mathbf{x})))$
Decoder: $p_\theta(\mathbf{x}|\mathbf{z})$
ELBO (a maximizar):
$$\mathcal{L}_{\text{VAE}} = \underbrace{\mathbb{E}_{q_\phi}[\ln p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{reconst.}} - \underbrace{\mathrm{KL}(q_\phi(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))}_{\text{regulariz.}}$$
KL gaussiano:
$\mathrm{KL}(\mathcal{N}(\boldsymbol{\mu},\mathrm{diag}(\boldsymbol{\sigma}^2))\|\mathcal{N}(\mathbf{0},\mathbf{I}))$
$= \frac12\sum_j(\mu_j^2+\sigma_j^2-\ln\sigma_j^2-1)$
Reparam.: $\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}) + \boldsymbol{\sigma}_\phi(\mathbf{x})\odot\boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon}\sim\mathcal{N}(\mathbf{0},\mathbf{I})$
(permite backprop a través del muestreo)

Information Theory

Entropía: $\mathbb{H}(p) = -\sum_i p_i\log p_i \ge 0$
KL divergence (no simétrica, ≥ 0):
$\mathrm{KL}(q\|p) = \sum_i q_i\ln\frac{q_i}{p_i} = \mathbb{E}_q\left[\ln\frac{q}{p}\right]$
$\mathrm{KL}(q\|p) = 0\Leftrightarrow q=p$; $\mathrm{KL}(q\|p)\ne\mathrm{KL}(p\|q)$
KL gaussianas ($p=\mathcal{N}(\boldsymbol{\mu}_1,\boldsymbol{\Sigma}_1)$, $q=\mathcal{N}(\boldsymbol{\mu}_0,\boldsymbol{\Sigma}_0)$):
$\mathrm{KL}(q\|p) =$
$\frac12\Big[\ln\frac{|\boldsymbol{\Sigma}_1|}{|\boldsymbol{\Sigma}_0|} + \mathrm{tr}(\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Sigma}_0) + (\boldsymbol{\mu}_1-\boldsymbol{\mu}_0)^\top\boldsymbol{\Sigma}_1^{-1}(\boldsymbol{\mu}_1-\boldsymbol{\mu}_0) - D$
Caso $p=\mathcal{N}(\mathbf{0},\mathbf{I})$:
KL $= \frac12\sum_j(\mu_j^2+\sigma_j^2-\ln\sigma_j^2-1)$

Métricas de Clustering

$\underset{\text{within}}{\text{WCSS}} = \sum_k\sum_{n:r_{nk}=1}\|\mathbf{x}_n-\boldsymbol{\mu}_k\|^2$ (↓ mejor)
$\underset{\text{between}}{\text{BCSS}} = \sum_k N_k\|\boldsymbol{\mu}_k - \bar{\mathbf{x}}\|^2$ (↑ mejor)
Silhouette $s(n)\in[-1,1]$: $a(n)$ = dist. intra, $b(n)$ = dist. al vecino más cercano
$s(n) = \frac{b(n)-a(n)}{\max(a(n),b(n))}$; $\bar s = \frac1N\sum_n s(n)$
Davies-Bouldin:
DB $= \frac1K\sum_k\max_{k'\ne k}\frac{s_k+s_{k'}}{d(\boldsymbol{\mu}_k,\boldsymbol{\mu}_{k'})}$
Elbow: graficar WCSS vs. K; codo = K óptimo.
BIC: $-2\ln\hat p(\mathbf{X}) + d_K\ln N$; ↓ mejor; selecciona K.
Mutual. Info (con labels):
$$I(S_k;Y) = \sum_i\sum_j p_{C_iY}(i,j)\log\frac{p_{C_iY}(i,j)}{p_{C_i}(i)\,p_Y(j)}$$
$p_{C_iY}(i,j) = \frac{|c_i\cap y_j|}{N}$; $I(X;Y)=0\Leftrightarrow X\perp Y$
$I(X;Y) = \mathbb{H}(X)-\mathbb{H}(X|Y) = \mathbb{H}(Y)-\mathbb{H}(Y|X)$
NMI (Normalized Mutual Information):
$$\mathrm{NMI}(S_k,Y) = \frac{2\,I(S_k;Y)}{\mathbb{H}(S_k)+\mathbb{H}(Y)}\in[0,1]\quad(\uparrow\text{ mejor})$$

Demo: EM-GMM - funcion Q y concavidad

E-step — $\mathcal{Q}(\mathbf{w},\mathbf{w}^{\text{old}})$:
$p(\mathbf{X},\mathbf{Z}|\mathbf{w}) = \prod_n\prod_k[\pi_k\mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)]^{z_{nk}}$
$\ln p(\mathbf{X},\mathbf{Z}|\mathbf{w}) =$
$\sum_n\sum_k z_{nk}[\ln\pi_k + \ln\mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)]$
$\mathcal{Q} = \sum_n\sum_k r_{nk}[\ln\pi_k - \frac12\ln|\boldsymbol{\Sigma}_k| - \frac12(\mathbf{x}_n-\boldsymbol{\mu}_k)^\top\boldsymbol{\Sigma}_k^{-1}(\mathbf{x}_n-\boldsymbol{\mu}_k)]$
M-step para $\boldsymbol{\mu}_k$: $\nabla_{\boldsymbol{\mu}_k}\mathcal{Q}=0$
$\sum_n r_{nk}\boldsymbol{\Sigma}_k^{-1}(\mathbf{x}_n-\boldsymbol{\mu}_k) = 0\Rightarrow\hat{\boldsymbol{\mu}}_k = \frac{\sum_n r_{nk}\mathbf{x}_n}{N_k}$
M-step para $\boldsymbol{\Sigma}_k$: $\nabla_{\Lambda_k}\mathcal{Q}=0$ con $\Lambda_k = \boldsymbol{\Sigma}_k^{-1}$:
$\frac{N_k}{2}\boldsymbol{\Sigma}_k - \frac12\sum_n r_{nk}(\mathbf{x}_n-\hat{\boldsymbol{\mu}}_k)(\mathbf{x}_n-\hat{\boldsymbol{\mu}}_k)^\top = 0$
$\Rightarrow\hat{\boldsymbol{\Sigma}}_k = \frac1{N_k}\sum_n r_{nk}(\mathbf{x}_n-\hat{\boldsymbol{\mu}}_k)(\mathbf{x}_n-\hat{\boldsymbol{\mu}}_k)^\top$
M-step para π_k (Lagrange $\sum\pi_k = 1$):
$\nabla_{\pi_k}[\mathcal{Q}+\nu(\sum\pi_k-1)] = 0\Rightarrow\hat\pi_k = \frac{N_k}{N}$
Concavidad de $\mathcal{Q}$: es una suma de $\ln\mathcal{N}$ ponderados; $\ln\mathcal{N}$ es cóncavo en $(\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)\Rightarrow\mathcal{Q}$ cóncava $\Rightarrow$ M-step es único. $\square$

Clustering+++

HAC (Hierarchical Agglomerative): Bottom-up: empieza con N clusters, fusiona iterativamente los más cercanos → dendrograma. No requiere K a priori.
Linkage: single (min dist), complete (max dist), average, Ward (min ΔWCSS).
DBSCAN: params ε, MinPts.
Core: $|\mathcal{N}_\varepsilon(\mathbf{x})|\ge$ MinPts; *Border*: vecino de core; *Noise*: resto.
Clusters = componentes conexas de cores + borders.

for $\mathbf{x}_n$ no visitado **do**
  if $|\mathcal{N}_\varepsilon(\mathbf{x}_n)|\ge$ MinPts **then** nuevo cluster; expandir($\mathbf{x}_n$)
  else noise

No requiere K; detecta formas arbitrarias; sensible a ε.
Spectral Clustering: Grafo de similitud $\mathbf{W}$; Laplaciano $\mathbf{L}=\mathbf{D}-\mathbf{W}$. Usar los K autovectores de menor λ de $\mathbf{L}$ como features; aplicar K-Means. Detecta clusters no convexos.
Mixture of Bernoullis ($\mathbf{x}\in\{0,1\}^D$):
$p(\mathbf{x}|\mathbf{w}) = \sum_k\pi_k\prod_d\mu_{kd}^{x_d}(1-\mu_{kd})^{1-x_d}$
EM igual que GMM; M-step: $\hat\mu_{kd} = \frac{\sum_n r_{nk}x_{nd}}{N_k}$

Álgebra Lineal

Traza y ciclo: $\mathrm{tr}(\mathbf{ABC}) = \mathrm{tr}(\mathbf{CAB}) = \mathrm{tr}(\mathbf{BCA})$
$\mathbf{x}^\top\mathbf{A}\mathbf{x} = \mathrm{tr}(\mathbf{A}\mathbf{x}\mathbf{x}^\top)$ (escalar → traza)
Gradientes matriciales clave:
$\nabla_{\mathbf{A}}\mathrm{tr}(\mathbf{AB}) = \mathbf{B}^\top$
$\nabla_{\mathbf{A}}\ln|\mathbf{A}| = \mathbf{A}^{-\top}$ (i.e. $\boldsymbol{\Sigma}^{-1}$ si $\boldsymbol{\Sigma}$ SDP)
$\nabla_{\mathbf{A}}(\mathbf{x}^\top\mathbf{A}^{-1}\mathbf{x}) = -\mathbf{A}^{-\top}\mathbf{x}\mathbf{x}^\top\mathbf{A}^{-\top}$
Identidades para $\boldsymbol{\Sigma}$ (sugeridas en ej. 20):
Si $\Lambda = \boldsymbol{\Sigma}^{-1}$: $\nabla_\Lambda\ln|\Lambda| = \Lambda^{-\top} = \boldsymbol{\Sigma}^\top$
$\nabla_\Lambda\mathrm{tr}(\Lambda\mathbf{S}) = \mathbf{S}^\top$
Proyección ortogonal: $\mathbf{P} = \hat{\mathbf{v}}\hat{\mathbf{v}}^\top$ (rango 1, idempotente)
$\|\mathbf{x}-\mathbf{P}\mathbf{x}\|^2 = \|\mathbf{x}\|^2 - (\mathbf{x}^\top\hat{\mathbf{v}})^2$
Matriz ortogonal R: $\mathbf{R}^\top\mathbf{R}=\mathbf{I}$, $|\det\mathbf{R}|=1$
$\|\mathbf{R}\mathbf{x}\|^2 = \mathbf{x}^\top\mathbf{R}^\top\mathbf{R}\mathbf{x} = \|\mathbf{x}\|^2$ (isometría)
Autovalores y traza/det:
$\mathrm{tr}(\mathbf{A}) = \sum_i\lambda_i$; $|\mathbf{A}| = \prod_i\lambda_i$
$\mathrm{tr}(\mathbf{AB}) = \mathrm{tr}(\mathbf{BA})$ incluso si $\mathbf{A},\mathbf{B}$ no cuadradas
Restricción con Lagrange:
$\max_{\hat{\mathbf{v}}}\hat{\mathbf{v}}^\top\hat{\boldsymbol{\Sigma}}\hat{\mathbf{v}}$ s.t. $\|\hat{\mathbf{v}}\|=1$
$\Rightarrow\mathcal{L} = \hat{\mathbf{v}}^\top\hat{\boldsymbol{\Sigma}}\hat{\mathbf{v}} - \lambda(\hat{\mathbf{v}}^\top\hat{\mathbf{v}}-1)$
$\Rightarrow\hat{\boldsymbol{\Sigma}}\hat{\mathbf{v}} = \lambda\hat{\mathbf{v}}$ (autoproblema)
$\Rightarrow$ máx. varianza $=\lambda_1$ con $\hat{\mathbf{v}}=\mathbf{u}_1$
Gaussiana multivariada $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\boldsymbol{\Sigma})$:
$$\frac{1}{(2\pi)^{D/2}|\boldsymbol{\Sigma}|^{1/2}}\exp\left(-\frac12(\mathbf{x}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$

Demo: PCA minimiza error de reconstruccion

Datos centrados ($\hat{\boldsymbol{\mu}}=0$). $\tilde x_i = z_i\hat{\mathbf{v}}$, $z_i = \mathbf{x}_i^\top\hat{\mathbf{v}}$.
MSE $= \frac1N\sum_i\|\mathbf{x}_i - z_i\hat{\mathbf{v}}\|^2$
$= \frac1N\sum_i\|\mathbf{x}_i\|^2 - \frac1N\sum_i(\mathbf{x}_i^\top\hat{\mathbf{v}})^2$
$=$ const $- \hat{\mathbf{v}}^\top\left(\frac1N\mathbf{X}^\top\mathbf{X}\right)\hat{\mathbf{v}}$
$\Rightarrow$ minimizar MSE ≡ maximizar $\hat{\mathbf{v}}^\top\hat{\boldsymbol{\Sigma}}\hat{\mathbf{v}}$.
Con restricción $\|\hat{\mathbf{v}}\|=1$ (Lagrange):
$\hat{\boldsymbol{\Sigma}}\hat{\mathbf{v}} = \lambda\hat{\mathbf{v}}$; la varianza $=\lambda$.
$\Rightarrow\hat{\mathbf{v}}^* = \mathbf{u}_1$ (autovector de $\lambda_{\max}$).

Demo: PCA invariante a roto-traslaciones

$\mathbf{x}_i' = R\mathbf{x}_i + \mathbf{b}$
Traslación: $\mathbf{x}_i' = \mathbf{x}+\mathbf{b} \implies \boldsymbol{\mu}' = \boldsymbol{\mu}+\mathbf{b}$.
$\tilde{\mathbf{x}} = \mathbf{x}_i' - \boldsymbol{\mu}' = \mathbf{x}_i + \mathbf{b} - (\boldsymbol{\mu}+\mathbf{b}) = \mathbf{x}_i - \boldsymbol{\mu}$.
Rotación: $\tilde{\mathbf{x}}_i' = \mathbf{R}\tilde{\mathbf{x}}_i$, $\mathbf{R}$ de rotación (⊥, det = 1).
$\hat{\boldsymbol{\Sigma}}' = \mathbf{R}\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top\mathbf{R}^\top = \mathbf{R}\hat{\boldsymbol{\Sigma}}\mathbf{R}^\top$
Si $\hat{\boldsymbol{\Sigma}}\mathbf{u}_l = \lambda_l\mathbf{u}_l \Rightarrow \hat{\boldsymbol{\Sigma}}'(\mathbf{R}\mathbf{u}_l) = \lambda_l(\mathbf{R}\mathbf{u}_l)$
Coords reducidas:
$z_l' = (\mathbf{R}\mathbf{u}_l)^\top(\mathbf{R}\tilde{\mathbf{x}}_i) = \mathbf{u}_l^\top\mathbf{R}^\top\mathbf{R}\tilde{\mathbf{x}}_i = z_l$.